

Question(s) for rec12 - Doubly Linked List class and Iterators

12. Consider the following code. What would be the result when attempting to build-and-run the program? Completely fill the circle next to your choice.

```
struct Node {
    int data;
    Node* next;
};

void foo(Node*& p) {                // Line A
    while (p != nullptr) {
        cout << p->data << ' ';
        p = p->next;
    }
    cout << endl;
}

int main() {
    Node* hp = nullptr;
    for (int i = 1; i <= 5; ++i) {
        hp = new Node{i, hp};
    }
    foo(hp);                        // Line B
    foo(hp);                        // Line C
}
```

- | | |
|--|---|
| ☐ 1 2 3 4 5
1 2 3 4 5 | ☐ The program fails to compile at line B |
| ☐ 1 2 3 4
1 2 3 4 | ☐ The program fails to compile at lines B and C |
| ☐ 5 4 3 2 1
5 4 3 2 1 | ☐ The program fails to compile for some other reason |
| ☐ 5 4 3 2
5 4 3 2 | ☐ The program compiles but does not generate any output when it runs. |
| ☐ The program fails to compile at line A | ☐ The program crashes when run. |
| | ☐ None of the above |